

# AI-Native Self-Healing Open-Radio Access Network (O-RAN) Network using Digital Twin

AI-Native Self-Healing O-RAN Network using Digital Twin proposes an autonomous network management framework that combines AI-based anomaly detection with a digital twin to predict, validate, and resolve RAN faults in a closed loop. The project targets improved reliability, faster recovery, and lower operational effort for future O-RAN deployments.

## Problem Statement

O-RAN networks are becoming more complex and distributed, which makes manual fault diagnosis and recovery slower and less reliable. Traditional static alarms or rule-based responses often react too late, require human intervention, and may not identify the true root cause of a problem. The project aims to solve this by combining AI and a digital twin to create a closed-loop self-healing mechanism that can sense, predict, decide, and verify recovery actions.

## Worklet objective

- Detect anomalies in O-RAN KPIs using AI-based monitoring.
- Predict faults before they cause major service disruption.
- Use a digital twin to simulate network conditions and validate healing actions safely.
- Automate corrective actions such as parameter tuning, rerouting, or recovery workflows.
- Compare the AI-native approach with traditional fault management methods.

## References

Research on **Closed-Loop Autonomous Network Healing Using Digital Twins**, which directly matches the project theme. The overview of **Digital Twin for O-RAN Towards 6G**, which explains why digital twins are useful for open and autonomous RANs.

Material on **AI in O-RAN** describing predictive maintenance, anomaly detection, and self-healing capabilities.

Work on **anomaly detection in Open RAN** showing how digital twins support controlled KPI streams, repeatable fault injection, and model training.

Broader AI-native autonomy discussions for O-RAN and 6G evolution.

<https://www.3gpp.org/technologies/5g-system-overview> (Basic)

## Core idea

Build a closed-loop system where a digital twin mirrors the live O-RAN environment, detects anomalies, predicts faults, and recommends or triggers healing actions before service impact grows. Research on O-RAN digital twins and AI-driven self-healing highlights this as a practical path toward autonomous RAN operations.

## Scope of work

The work should cover anomaly detection, root-cause analysis, fault prediction, and automated recovery in an O-RAN environment. The digital twin can be used to emulate network states, inject faults, test mitigation actions safely, and train AI models using realistic KPI streams. A strong PRISM scope also includes comparison against rule-based healing and evaluation under different fault scenarios such as configuration errors, traffic overload, and node/link degradation.

## Expected Outcome

- A conceptual architecture for AI-native self-healing in O-RAN.
- A digital-twin-based testbed for fault injection and recovery validation.
- AI models for anomaly detection or predictive fault identification.
- Faster fault recovery and reduced downtime compared with manual or rule-based methods.
- Evaluation results showing improvements in availability, recovery time, and operational efficiency.

| M1: 0-8 days                                                                                                                                                                                                           | M2, & M3 (10-22 days)                                                                                                                                                                   | M4, M5, M6 (24-36 days)                                                                                                                                                                                                                                                                                 | M6 (40-50 days)                                                                                                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Problem study and Literature Survey</b> Define O-RAN self-healing requirements and digital twin use cases.<br><b>System Architecture design</b> Finalize data flow between live O-RAN, AI engine, and digital twin. | <b>M2: KPI and fault taxonomy</b> Identify KPIs, anomaly types, and healing targets.<br><b>M3: Digital twin setup</b> Build or simulate the O-RAN twin and fault injection environment. | <b>M4: AI model development</b> Implement anomaly detection or predictive fault models.<br><b>M5: Healing logic integration</b> Add decision engine for automated or assisted recovery actions.<br><b>M6: Testing and evaluation</b> Measure detection accuracy, recovery time, and service continuity. | <b>M6: Paper &amp; IP Draft</b> (Publication + Patent disclosure) Final documentation Prepare results, findings, and future enhancement ideas |

## Students:

Tanmaya Kumar bl.en.u4eac23058@bl.students.amrita.edu

Raghu Ram K bl.sc.u4cse24142@bl.students.amrita.edu

Munipalle Jaswanth Kumar bl.sc.u4cse24130@bl.students.amrita.edu

## Mentors:

Bikas Singh (NovaThink Tech)

Navin Kumar ([k\\_Navin@blr.amrita.edu](mailto:Navin@blr.amrita.edu))  
9019930325

# Open Fronthaul S-Plane Timing Security

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Sub-scope of the AI-Native Self-Healing O-RAN Network using a Digital Twin worklet

## TEAM

Tanmaya Kumar · Raghu Ram K · Munipalle Jaswanth Kumar

## MENTORS

Mr. Bikas Singh (NovaThink Tech)  
Dr. Navin Kumar (Amrita Vishwa Vidyapeetham)

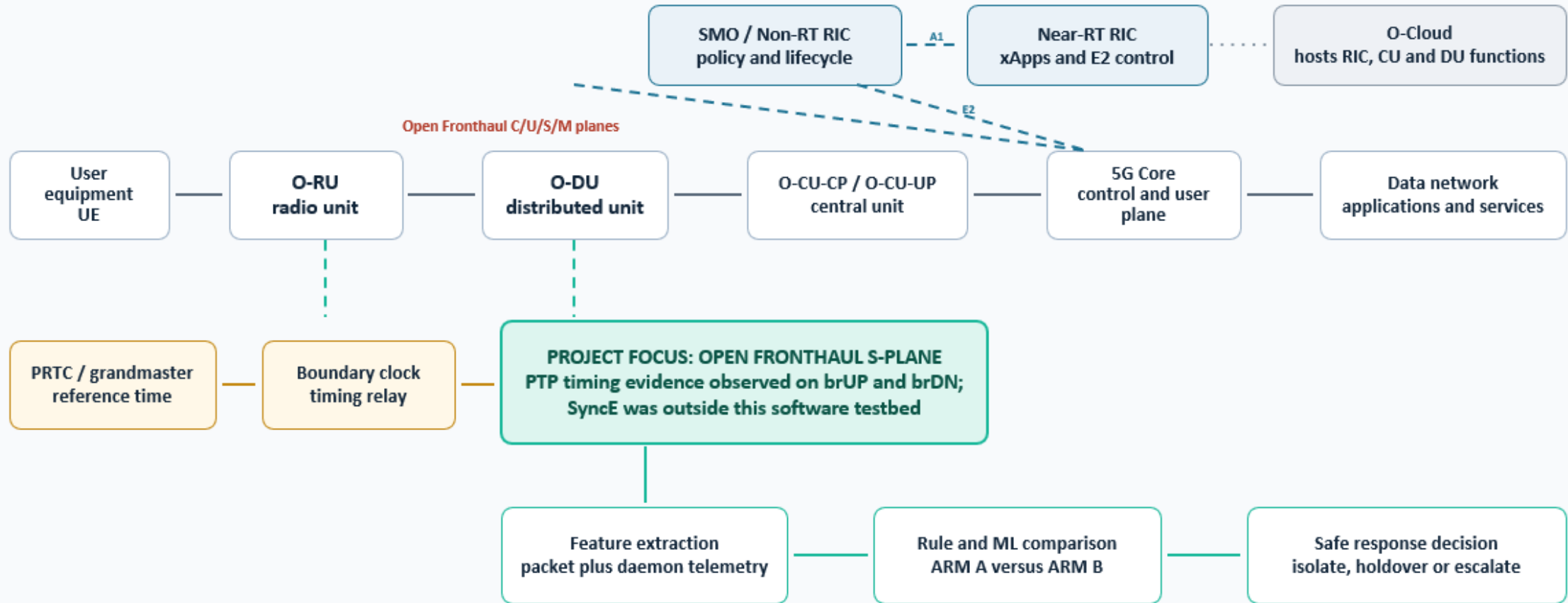
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Amrita School of Engineering, Bengaluru · Samsung PRISM

168 live on-wire runs · 14 fault scenarios · 12 randomised replicates

# End-to-End O-RAN Architecture and S-Plane Project Boundary

O-RAN Alliance architecture description (ETSI TS 103 982 PAS) | O-RAN.WG4.CUS.0 Control, User and Synchronization Plane Specification | 3GPP TS 38.401 NG-RAN architecture



Validated here: live packet-path execution, extracted timing evidence and classification. Remaining scope: digital-twin orchestration, fault prediction, automated recovery and outcome measurement.

# Open Fronthaul Timing Requirements and Security Exposure

O-RAN.WG4.CUS.0 timing categories (via ATIS/WSTS O-RAN Fundamentals tutorial, Armstrong 2022) | ITU-T G.8271.1 network time-error budget | Groen et al., TIMESAFE, ACM ToPS 28(5), 2025, doi:10.1145/3775060

|                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>130 ns</div> <div>Relative time alignment error between radio units in a cluster</div> <div>Timing Category A; implies <math>\approx \pm 65</math> ns per O-RU against a common reference under equal allocation</div> <div>O-RAN.WG4.CUS.0</div> | <div><math>\pm 1.5 \mu\text{s}</math></div> <div>End-application absolute time error, O-RU to primary reference</div> <div>Total budget at reference point E; the network limit at reference point C is 1100 ns</div> <div>ITU-T G.8271.1</div> | <div><math>\approx 2</math> s</div> <div>Observed time from attack onset to O-RU crash in one published study</div> <div>The O-RU crashed and required a manual reboot, with switch PTP ports in dynamic role</div> <div>TIMESAFE, ACM ToPS 2025</div> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**WHY THIS IS A SECURITY SURFACE, NOT ONLY AN ENGINEERING CONSTRAINT**

Time is distributed across the fronthaul as ordinary Ethernet frames carrying the IEEE 1588 protocol, not over a dedicated physical channel. Any device able to place frames on that path can therefore advertise itself as a time source, forge timing content, or withhold legitimate messages. Loss of a shared time reference degrades radio coordination, so the S-plane is simultaneously a timing-performance boundary and an attack surface.

**Figure basis**      The 130 ns and  $\pm 1.5 \mu\text{s}$  values derive from 3GPP air-interface requirements re-expressed by O-RAN and ITU-T. The  $\approx 2$  s figure is a single observed experimental result, not a standardised limit.

# Attack–Fault Ambiguity and Sub-Scope Selection

IEEE 1588-2019 clock states and timePropertiesDS | ITU-T G.8275.1 clockClass ladder (6 locked, 7 holdover in spec) | O-RAN WG11 Threat Model, ETSI TR 104 106 V3.0.0

**BENIGN FAULT**

## The clock is wrong

Example: a GNSS outage at the grandmaster, declared through clockClass 6 → 7.

**Correct response — ride it out in holdover.**

Isolating the node here creates an outage that was not going to occur.

**ATTACK**

## The clock is wrong

Example: a rogue master injecting false time through forged Announce messages.

**Correct response — isolate the source immediately.**

Waiting in holdover allows the attacker to keep steering the clock.

**Identical observable symptom · opposite correct response · the decision cannot be made from the symptom alone**

RATIONALE FOR SELECTING THE S-PLANE SUB-SCOPE

**01 Highest consequence**

A timing failure can become a service failure within seconds, leaving little time for a safe decision.

**02 Genuine ambiguity**

Selected benign faults and attacks produce overlapping symptoms; the decision needs provisioned context.

**03 Constrained by standards**

IEEE 1588-2019 and ITU-T G.8275.1 fix wire-level legality, giving objective decision criteria.

**04 Reproducible in software**

Real PTP daemons can be executed and attacked on a virtual wire without a proprietary RAN stack.

**The sub-scope was selected because it is the point where a detection label alone is insufficient: the same symptom requires opposite recovery actions, and standards supply objective criteria for separating them.**

# Research Foundation for S-Plane Timing Assurance

Groen et al., TIMESAFE, ACM ToPS 28(5), 2025, doi:10.1145/3775060 | Bonati et al., Colosseum, IEEE OJ-COMS, 2024, doi:10.1109/OJCOMS.2024.3447472  
Digital Twin for O-RAN, IEEE Communications Magazine, 2024, doi:10.1109/MCOM.003.2400016 | OpenRAN Gym, Computer Networks, 2023 | TSN Digital Twin, EURASIP JIS, 2025, doi:10.1186/s13635-025-00213-7

| SOURCE                                    | WHAT THE LITERATURE ESTABLISHES                                              | BOUNDARY LEFT OPEN                                                             |
|-------------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>TIMESAFE · ACM ToPS 2025</b>           | PTP attacks disrupt O-RAN fronthaul and are detectable from packet evidence. | Detection was not tested against a matched set of benign timing faults.        |
| <b>Digital Twin for O-RAN · IEEE 2024</b> | A RAN twin supports testing, optimization and safer control decisions.       | The architecture does not validate S-plane attack-versus-fault discrimination. |
| <b>Colosseum · IEEE 2024</b>              | Repeatable Open RAN twins reproduce realistic experimental conditions.       | The platform does not address telecom PTP legality or recovery choice.         |
| <b>OpenRAN Gym · Elsevier 2023</b>        | O-RAN AI/ML requires controlled data collection and reproducible evaluation. | It does not test whether an S-plane model beats rules or a constant predictor. |
| <b>TSN Digital Twin · Springer 2025</b>   | A timing-aware twin supports structured security testing.                    | The study does not combine O-RAN PTP, benign lookalikes and held-out controls. |

Prior work establishes the threat, the AI testbed direction and the digital-twin direction. It does not resolve the operational decision between malicious timing behaviour and a benign timing fault presenting the same symptom.

# Research Gaps Behind the Experimental Design

Federated Continual Learning for O-RAN Anomaly Detection, IEEE WCNC 2024 | Adversarial ML Threat Analysis in O-RAN, JNCA 2024, doi:10.1016/j.jnca.2024.104090  
O-RAN xApps: Survey and Research Challenges, Computer Networks 2025 | Anomaly Detection for xApp and E2 Threats, IEEE 2025

01

## Attack and fault ambiguity

Prior detectors separate attack from normal operation. Operators must also separate attacks from real clock, path and configuration faults.

02

## Unverified AI claims

Reported model accuracy can appear strong without a held-out split, a rule comparator and a trivial constant baseline.

03

## Protocol evidence separated from ML

Standards checks and statistical models are usually presented as alternatives rather than evaluated on the same executed runs.

04

## Detection separated from safe response

A label alone does not determine whether to isolate a source, enter holdover or escalate to an operator.

### RESULTING STUDY DESIGN

#### Matched scenarios

Attacks and benign lookalikes execute through the same packet path.

#### Two decision arms

ARM A applies standards legality plus operator context. ARM B applies Random Forest with an Isolation Forest novelty layer.

#### Held-out evaluation

Replicates 9–12 remain outside training for the 56-run final test.

#### Trivial control

An always-BENIGN predictor establishes whether ARM B adds information.

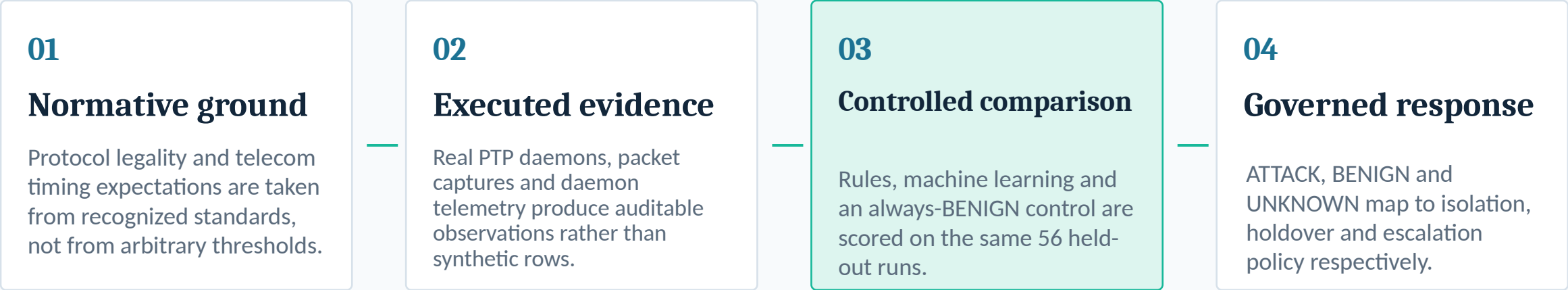
#### Explicit abstention

Ambiguous evidence escalates instead of forcing an unsafe answer.

Gap addressed: the reviewed source set did not combine S-plane benign lookalikes, standards-aware rules, a machine-learning arm, explicit abstention and a constant baseline within one held-out campaign.

# Standards-Grounded Positioning of the Contribution

Normative basis: IEEE 1588-2019, ITU-T G.8275.1, O-RAN.WG4.CUS.0, O-RAN WG11 (ETSI TR 104 106 V3.0.0) | Prior-art boundary: TIMESAFE, Colosseum, OpenRAN Gym, digital-twin literature



**DEMONSTRATED IN THIS CAMPAIGN**

- Live PTP packet-path execution
- Held-out attack and benign-fault classification
- ARM A versus ARM B versus constant control
- Explicit abstention with evidence provenance

**NOT YET DEMONSTRATED**

- Digital-twin action verification
- Executed isolate or holdover recovery
- Measured MTTR and availability improvement
- An ML model exceeding transparent baselines

**Contribution: a standards-aware, controlled S-plane benchmark that tests whether a learned model adds decision value before it is granted authority over recovery.**

# Telecom Standards and Profile Grounding

Normative basis: IEEE Std 1588-2019 | O-RAN.WG4.CUS.0 | O-RAN WG11 Security Threat Modeling (ETSI TR 104 106 V3.0.0) | ITU-T G.8275.1  
Timing performance context: ITU-T G.8273.2 | Frequency synchronization context: ITU-T G.8262 (SyncE, not implemented here)

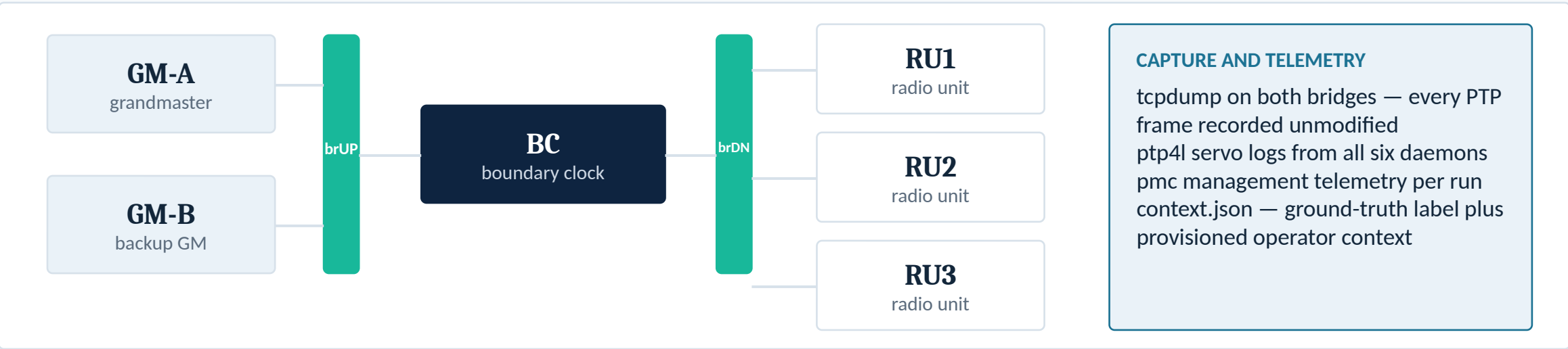
| AUTHORITY       | WHAT IT DEFINES                                                                | HOW THIS PROJECT USES IT                                                           |
|-----------------|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| IEEE 1588-2019  | Legal PTP message fields, data sets, clock behaviour and leap-second handling. | ARM A checks field legality, sequence continuity and timePropertiesDS consistency. |
| O-RAN.WG4.CUS.0 | Open Fronthaul C/U/S-plane architecture and synchronization configurations.    | Defines the O-DU-O-RU S-plane boundary used by the experiment.                     |
| O-RAN WG11      | Open Fronthaul threat model, including malicious timing-source behaviour.      | Supplies the threat identifiers that label the attack scenarios.                   |
| ITU-T G.8275.1  | Telecom PTP profile for phase and time distribution with full timing support.  | Fixes domain, message rates, transport and the alternate BMCA used by the testbed. |
| ITU-T G.8273.2  | Timing characteristics and limits for telecom boundary clocks.                 | Provides performance context for interpreting timing-quality evidence.             |

**Scope disclosure**      The testbed exercised PTP only; SyncE is a physical-layer function and was not implemented. ITU-T recommendations are paywalled, so several values were confirmed against authoritative secondary sources, as recorded in the project workbooks.

# Open Fronthaul Timing Testbed Architecture

linuxptp v4.0 (released 2023-06-09), reference implementation of IEEE 1588-2019 and ITU-T G.8275.1 | Linux network namespaces, veth pairs and software bridges | tcpdump packet capture

Six genuine linuxptp 4.0 daemons in isolated Linux network namespaces across two bridged segments. All protocol behaviour — BMCA election, servo control, message cadence — is executed by real daemons, not simulated.



**ITU-T G.8275.1 profile as configured**  
domain 24 · priority1 128 · Sync 16/s (logInterval -4) · Announce 8/s (logInterval -3)

**Evidence boundary**  
No independent clock-error instrument was used; synchronization impact is inferred from protocol and servo telemetry, not measured against a reference clock.

Faults are produced by live packet-path mechanisms — competing daemons, injected frames, replayed captures and link or queue manipulation — never by editing rows in a dataset.

# Timing Configuration and Verified Platform Constraints

ITU-T G.8275.1 telecom profile (values cross-checked against Cisco and Nokia profile documentation and linuxptp configs/G.8275.1.cfg) | Constraints confirmed by direct query of the running system

| PARAMETER     | CONFIGURED VALUE               | BASIS                                                                       |
|---------------|--------------------------------|-----------------------------------------------------------------------------|
| Profile       | ITU-T G.8275.1 telecom profile | Full timing support from the network                                        |
| domainNumber  | 24                             | G.8275.1 permitted range 24-43                                              |
| priority1     | 128 (fixed)                    | G.8275.1 fixes the value and the alternate BMCA excludes it from comparison |
| Sync rate     | 16 / s (logInterval -4)        | G.8275.1                                                                    |
| Announce rate | 8 / s (logInterval -3)         | G.8275.1                                                                    |
| BMCA          | dataset_comparison G.8275.x    | Alternate BMCA, not the IEEE 1588 default                                   |
| Transport     | L2 multicast 01-1B-19-00-00-00 | Forwardable address, permitted by G.8275.1 (linuxptp default)               |
| Timestamping  | software                       | Platform limit — see constraints                                            |

VERIFIED PLATFORM CONSTRAINTS

! No PTP hardware clock

Is /dev/ptp\* returns empty. Software timestamping only: a microsecond-scale noise floor against a nanosecond-scale target.

! No loadable kernel modules

modprobe absent and sch\_netem unavailable, so congestion uses tbfb.

! One shared oscillator

All namespaces share the host clock, so clocks cannot drift apart.

! No GNSS receiver

GNSS spoofing and jamming cannot be produced at all.

**Effect on scope** These constraints set the ceiling on every result that follows: five of the sixteen catalogued fault classes cannot be produced here, and no claim is made about absolute time-error magnitude.

Every configured value was read back from the running system and traced to the governing standard or, where the normative text is paywalled, to an authoritative secondary source.

# Data Flow from Packet Capture to Verdict

Deep extractor: 56 fields per PTP frame | Field definitions from IEEE 1588-2019 message and data-set clauses | Conformance flags derived from ITU-T G.8275.1 profile constraints

01

## The wire

Real PTP frames produced by six linuxptp daemons on two bridged segments

›

02

## Capture

tcpdump records every frame on brUP and brDN, unmodified

›

03

## Deep extract

56 protocol and conformance fields per packet; fields that cannot be observed are written EMPTY

›

04

## Decision rule

Standards legality plus provisioned operator context, applied as ordered clauses

›

05

## Verdict

ATTACK · BENIGN · UNKNOWN, with the triggering clause recorded

### INTEGRITY RULE CARRIED THROUGH THE WHOLE PIPELINE

## A value that cannot be measured is written EMPTY — never zero.

A zero offset reads as perfect synchronization. An earlier build wrote zero for unobserved fields and consequently reported a blind node as healthy. Nothing is treated as healthy without positive proof of observation, and this rule is applied at extraction, at feature construction and at scoring.

The same extracted evidence feeds both decision arms, so the rule-based and machine-learning comparison is performed on identical executed runs rather than on separately prepared datasets.

# Experimental Dataset Composition

Packaged archive: splane\_campaign\_CORRECTED\_2026-09-20.tgz, 1096 files, sha256 6149b4fb15940cdac694ba8666d97041b4eb62d5523d2631c1edf96d531e4ed9 | Run inventory enumerated in EVALUATION\_V4.json

168

live on-wire runs

14

fault scenarios

12

randomised replicates each

1,451,909

PTP packet records extracted

56

fields per packet

WHAT CONSTITUTES ONE RUN

Two bridge captures

every PTP frame on brUP and brDN, unmodified

One deep-extract CSV

56 fields per packet; unobservable fields written EMPTY

Six daemon servo logs

offset, path delay and frequency per node

One pmc telemetry file

management data set snapshots during the run

One context.json

ground-truth class, fault id, allow-list, maintenance state

Three verdict files

base rule, v2 and v3 scored on the same capture

MEASURABLE AND UNAVAILABLE EVIDENCE

Recorded for every run

Packet header, flag, correction, identity, sequence and Announce data-set fields; derived G.8275.1 conformance flags; per-node servo telemetry.

Structurally unavailable

GNSS status, SyncE quality level, oscillator frequency error and absolute clock error. No hardware exists in this testbed to produce them, so the corresponding columns are EMPTY rather than defaulted.

Partially available

pmc management telemetry is absent for 48 of the 168 runs (all B3, C1, C2 and C3 replicates); this is recorded rather than imputed.

**RECORD COUNT VARIES WITH THE FAULT, AND THE VARIATION IS ITSELF EVIDENCE**

C1 interception averages 2,360 records per run because the attack starves the link · baseline averages 6,021 · A5 flooding averages 38,319 because the attack adds traffic. Record volume is therefore a discriminating feature, not a data-quality artefact.

Archive contents

Roughly 222 MB of source PCAPs were captured across the campaign. The packaged review archive retains the extracted CSVs, verdicts, context files, daemon logs, scripts and freeze records; it does not retain the source PCAPs.

# Fault Injection Mechanisms on the Live Packet Path

IEEE 1588-2019: message formats, sequenceId continuity, timePropertiesDS §8.2.4, leap-second handling §9.4 | O-RAN WG11 security test identifiers (ETSI TS 104 105) | Scapy, tcpreplay, Linux tc

## Additional real PTP daemons

A1 • A8

A new namespace runs a genuine ptp4l instance with an identity outside the provisioned allow-list and a superior data set, then competes for election under the G.8275.1 alternate BMCA. The attack is won on protocol merit, not by editing a capture.

## Crafted frame injection

A2 • C2 • C3

Hand-built frames placed on the RU segment: forged two-step timing messages; malformed frames using versionPTP 3, a reserved message type and a short length; Announce frames with leap61 set and the UTC offset falsified.

## Capture and replay

A3

tcpdump captures live frames mid-run and tcpreplay retransmits them, producing genuinely stale origin timestamps and reused sequence identifiers rather than synthetic duplicates.

## Link and queue manipulation

A5 • C1 • B3 • B7

High-rate flooding; sustained blackholing of the boundary clock's downstream port; a tbf bottleneck carrying competing non-PTP traffic; and bouncing an RU link during a maintenance window.

An attack and its benign twin frequently share the same packet-path mechanism. Killing a grandmaster is identical in B2 and B\_unplanned\_failover; only the provisioned maintenance context differs. The context is therefore part of the evidence, not metadata.

# Attack Scenarios: Identity and Message Manipulation

IEEE 1588-2019 Announce, Sync and Follow\_Up message clauses and sequenceId continuity | ITU-T G.8275.1 alternate BMCA and message-rate cadence | O-RAN WG11 threat identifiers, ETSI TR 104 106 V3.0.0

| SCENARIO                                                             | INJECTED CONDITION                                                                                                                              | OBSERVABLE EFFECT                                                                                                        | EXPECTED | RESULT           |
|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|----------|------------------|
| <b>A1</b><br>Rogue grandmaster / BMCA spoof<br><br>T-SPLANE-02 / -03 | A second real ptp4l instance with an identity outside the provisioned allow-list and a superior priority2 value competes in the election.       | The rogue clock wins the alternate BMCA and becomes the advertised time source for the downstream segment.               | ATTACK   | 12/12<br>correct |
| <b>A2</b><br>Sync / Follow_Up spoofing<br><br>IEEE 1588-2019         | Forged timing messages from an unallow-listed source identity, sent in bursts, with implausible origin timestamps and no accompanying Announce. | Timing content conflicts with the elected grandmaster while the source never advertises itself legitimately.             | ATTACK   | 12/12<br>correct |
| <b>A3</b><br>Replay<br><br>IEEE 1588-2019 sequenceId                 | Live frames captured mid-run and retransmitted with tcpreplay, carrying genuinely stale origin timestamps and reused sequence identifiers.      | Sequence continuity regresses and timestamps move backwards relative to the active stream.                               | ATTACK   | 12/12<br>correct |
| <b>A5</b><br>DoS / PTP flooding<br><br>T-SPLANE-01                   | High-rate PTP traffic injected on the RU segment in bursts of 300 frames, from a source outside the allow-list.                                 | Message rate far exceeds the profile cadence; record volume rises to roughly 38,000 per run against a baseline of 6,000. | ATTACK   | 12/12<br>correct |

All four scenarios were detected by the pre-frozen base rule at 12/12, before any rule revision. Detection rests on protocol legality and allow-list membership, not on the magnitude of the timing error produced.

# Attack Scenarios: Relay, Interception and Field Abuse

O-RAN WG11 security test identifiers 11.1.5.3.1 and 24.2.1.2 (ETSI TS 104 105) | IEEE 1588-2019 stepsRemoved, timePropertiesDS §8.2.4 and leap-second handling §9.4 | ETSI TR 104 106 V3.0.0

| SCENARIO                                                           | INJECTED CONDITION                                                                                                                     | OBSERVABLE EFFECT                                                                                                              | EXPECTED | RESULT           |
|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|----------|------------------|
| <b>A8</b><br>Rogue boundary clock<br><br>T-SPLANE-03               | A two-port ptp4l instance slaves to the legitimate grandmaster upstream and re-advertises downstream with stepsRemoved incremented.    | An unauthorised clock is inserted into the timing path while still relaying genuine time, so content alone appears valid.      | ATTACK   | 12/12<br>correct |
| <b>C1</b><br>Selective interception and removal<br><br>T-SPLANE-04 | The boundary clock's downstream port is blackholed for a sustained fraction of the observation window, withholding Sync and Announce.  | Radio units receive messages below the rate the boundary clock itself declares; delivered-to-declared ratio falls toward 0.37. | ATTACK   | 11/12<br>correct |
| <b>C2</b><br>Malformed / fuzzed frames<br><br>WG11 24.2.1.2        | Frames injected with versionPTP 3, a reserved message type, and a message length shorter than the protocol permits.                    | Frames are structurally illegal under IEEE 1588-2019 regardless of their timing content.                                       | ATTACK   | 12/12<br>correct |
| <b>C3</b><br>Whole-second field abuse<br><br>IEEE 1588-2019 §8.2.4 | Announce frames carrying leap61 asserted, currentUtcOffset falsified to 0 against a true value of 37, and traceability flags stripped. | Time properties become internally inconsistent, which would shift the applied UTC offset by a whole second.                    | ATTACK   | 12/12<br>correct |

Note

The pre-frozen base rule returned BENIGN on C1, C2 and C3 at 0/12 each. These three coverage gaps were closed by a disclosed post-campaign rule revision, evaluated on the same captures and reported separately from the pre-frozen result.

# Benign Fault and Operational-Change Scenarios

ITU-T G.8275.1 clockClass ladder and stepsRemoved semantics | IEEE 1588-2019 BMCA re-election behaviour | Provisioned operator context: allow-list, maintenance-window state, expected boundary-clock identity

| SCENARIO                                                                 | INJECTED CONDITION                                                                                                            | OBSERVABLE EFFECT                                                                                                                           | EXPECTED      | RESULT                      |
|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------------------------|
| <b>baseline</b><br>Healthy control<br>—                                  | All six daemons nominal: grandmaster A elected, relayed through the boundary clock to three radio units.                      | Stable offset and path delay; message cadence matches the declared profile rates.                                                           | <b>BENIGN</b> | <b>12/12</b><br>correct     |
| <b>B2</b><br>Planned grandmaster changeover<br>ITU-T G.8275.1            | Grandmaster A withdrawn during an open maintenance window; grandmaster B, which is on the provisioned allow-list, is elected. | A grandmaster change identical in packet terms to A1 — the distinguishing evidence is allow-list membership plus maintenance state.         | <b>BENIGN</b> | <b>12/12</b><br>correct     |
| <b>B3</b><br>Packet delay variation and congestion<br>ITU-T G.8261       | A 1 Mbit tbf bottleneck on the boundary-clock to radio-unit path carrying competing non-PTP background traffic.               | Real queueing delay: Sync packet delay variation rises from 0.07 ms to between 5.6 and 857 ms, with no packet loss and no protocol anomaly. | <b>BENIGN</b> | <b>11/12</b><br>correct     |
| <b>B7</b><br>Topology reconfiguration<br>IEEE 1588-2019                  | A radio-unit link is bounced and re-homed during an open maintenance window.                                                  | stepsRemoved and measured path delay change consistently with the new topology, with no illegal field values.                               | <b>BENIGN</b> | <b>12/12</b><br>correct     |
| <b>B_bc_repl</b><br>Planned boundary-clock replacement<br>ITU-T G.8275.1 | A provisioned replacement boundary clock is swapped in during maintenance; its identity is recorded in the operator context.  | A new clock identity appears in the timing path relaying the genuine grandmaster — structurally identical to the A8 rogue relay.            | <b>BENIGN</b> | <b>0/12</b><br>open failure |

Note

Four of five benign scenarios were classified correctly. The boundary-clock replacement is the single systematic failure: the decision context carries one expected boundary-clock identity, so a legitimate replacement is read as an unauthorised relay.

# Ambiguous Scenario and Catalogue Coverage

Sixteen-class S-plane fault catalogue maintained in the project parameter matrix (144 parameters × 16 fault classes) | Threat identifiers from ETSI TR 104 106 V3.0.0 | Coverage assessed against verified platform constraints

**B\_unplanned\_failover • DELIBERATELY AMBIGUOUS**  
**Grandmaster A fails with no maintenance window open; provisioned grandmaster B is elected.**  
From packet evidence alone this is indistinguishable from an attack that suppressed the grandmaster. Both produce an unplanned election to a legitimate backup. The decision rule returns UNKNOWN and escalates rather than forcing a binary answer.

EXPECTED UNKNOWN

**12 / 12 correct**

Abstention is scored on its own axis, not as a detection success.

SIXTEEN-CLASS CATALOGUE COVERAGE

|                                                                                                    |                                                                                               |                                                                                              |                                                                                              |
|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| <div><b>A1</b> <span>SW</span> <span>TESTED</span></div> <div>Rogue grandmaster / BMCA spoof</div> | <div><b>A2</b> <span>SW</span> <span>TESTED</span></div> <div>Sync / Follow_Up spoofing</div> | <div><b>A3</b> <span>SW</span> <span>TESTED</span></div> <div>Replay</div>                   | <div><b>A4</b> <span>SW*</span> <span>not run</span></div> <div>Delay (time) attack</div>    |
| <div><b>A5</b> <span>SW</span> <span>TESTED</span></div> <div>DoS / PTP flooding</div>             | <div><b>A6</b> <span>HW</span> <span>blocked</span></div> <div>GNSS spoofing</div>            | <div><b>A7</b> <span>HW</span> <span>blocked</span></div> <div>GNSS jamming</div>            | <div><b>A8</b> <span>SW*</span> <span>TESTED</span></div> <div>Rogue boundary clock</div>    |
| <div><b>B1</b> <span>HW</span> <span>blocked</span></div> <div>GNSS holdover</div>                 | <div><b>B2</b> <span>SW</span> <span>TESTED</span></div> <div>Planned GM changeover</div>     | <div><b>B3</b> <span>SW</span> <span>TESTED</span></div> <div>PDV / congestion</div>         | <div><b>B4</b> <span>HW</span> <span>blocked</span></div> <div>SyncE / EEC degradation</div> |
| <div><b>B5</b> <span>SW*</span> <span>not run</span></div> <div>Static path asymmetry</div>        | <div><b>B6</b> <span>HW</span> <span>blocked</span></div> <div>Oscillator drift</div>         | <div><b>B7</b> <span>SW</span> <span>TESTED</span></div> <div>Topology reconfiguration</div> | <div><b>B8</b> <span>SW</span> <span>not run</span></div> <div>Measurement noise floor</div> |

Eight of sixteen catalogued classes were executed. Five require physical timing hardware and cannot be produced here. Three have software detection logic only, where physical impact would need hardware to measure. SW\* denotes detection logic without physical-impact measurement.

# Classification Logic and Verdict Assignment

IEEE 1588-2019 field legality, data-set consistency and sequence continuity | ITU-T G.8275.1 profile constraints and clockClass ladder | Provisioned operator context supplied per run in context.json

**BENIGN FAULT SIGNATURE**

Degradation that is self-consistent with a declared, standards-compliant state change: clockClass 6 to 7 for a grandmaster entering holdover within specification, stepsRemoved changing consistently with a new topology, or delay variation rising without any field becoming illegal.

The declaration and the observation agree.

**ATTACK SIGNATURE**

Evidence that is illegal or unauthorised: forged Announce fields, non-monotonic sequence identifiers, structurally invalid frames, a delivered message rate below the rate the source itself declares, or a superior clock whose identity is outside the provisioned allow-list.

The declaration and the observation disagree.

VERDICT ASSIGNMENT AND RECOMMENDED RESPONSE

**ATTACK**

Protocol-inconsistent or unauthorised evidence present

ISOLATE

**BENIGN**

Only provisioned clocks, no legality violation

TOLERATE / HOLDOVER

**UNKNOWN**

Packet evidence cannot distinguish intent

ESCALATE

Recommended response labels only. No corrective action was executed on this testbed.

The worked discriminator: a grandmaster suddenly changes. It could be B2, a planned failover, or A1, BMCA spoofing. The decisive test is whether the new clock identity appears on the provisioned allow-list and whether a maintenance window is open. Both scenarios were classified correctly at 12/12.

On the deliberately ambiguous unplanned failover the rule returned UNKNOWN on 12 of 12 runs. A BENIGN call there would be an unsafe over-claim and an ATTACK call a false alarm; both are treated as failures.

# Campaign Design and Reproducibility Controls

Decision rule SHA-256 hashed before execution (FROZEN.json) | Per-replicate randomisation seeded from the replicate number | Pre-registered expected verdicts | Full run inventory in EVALUATION\_V4.json

168

live on-wire runs

14

scenarios

12

randomised replicates each

3

rules scored on identical captures

|                              |                                                                                                                                                                                      |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rule frozen before execution | The base decision rule was SHA-256 hashed before any run started, so it could not be tuned to the outcome. The hash is recorded in the packaged archive.                             |
| Revision history disclosed   | Rule v3 was written after v2 false positives were observed, then re-hashed before scoring. It is reported as a transparent post-observation revision, not as independent validation. |
| Pre-registered expectations  | Each scenario's expected verdict was written down in advance. Sensitivity, specificity and abstention are scored on separate axes rather than pooled into one figure.                |
| Per-replicate randomisation  | Attacker identity, priority, burst size, inter-frame gap and onset are drawn reproducibly from the replicate number, so repeats test generalisation rather than repetition.          |
| Per-scenario reporting       | Per-scenario recall is the primary metric. Pooled figures move with how many runs of each scenario exist, so they are labelled descriptive only.                                     |

Design limit

The three C-series scenarios draw from only four to five distinct randomised parameter values, so twelve replicates are twelve genuine captures but not twelve independent attack designs.

# Pipeline Audit and Corrective Reruns

Independent audit of the campaign apparatus, 2026-09-20 | Defect evidence and corrective actions recorded in AUDIT\_AND\_CORRECTIONS | All affected runs re-executed before any result in this deck was computed

A cross-check audit of the campaign apparatus identified four defects. Two of them had produced invalid data. Every affected run was re-executed, and the results reported later in this deck are computed only from the corrected campaign.

|                                                                                                                                                                                                                                                                  |                                                                                                                                                                  |                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <b>Thirty-six replicates were three captures copied twelve times</b><br>A runner script was created without the execute bit. The resulting exit code 126 was silenced by output redirection, and a stale capture then satisfied the packet-count validity guard. | Proof: one distinct SHA-256 across twelve supposed replicates, and completion in under one second where a genuine run takes about 48 seconds.                    | FIXED • re-run  |
| <b>A benign scenario never injected anything</b><br>This kernel has no sch_netem qdisc, and the failure was swallowed by a shell fallback. B3 congestion was a baseline run carrying a congestion label.                                                         | Proof: B3 packet delay variation was statistically identical to baseline and to the untouched control node. Rebuilt with a tbf bottleneck and competing traffic. | FIXED • rebuilt |
| <b>A detector produced false positives on upstream captures</b><br>Rule v2's rate-starvation clause had no minimum-sample or transient guard, so a boundary clock emitting three frames during BMCA start-up read as rate-starved.                               | Effect: it turned a benign failover into an ATTACK and destroyed a correct abstention. Rule v3 adds a minimum-frame floor and a persistence requirement.         | FIXED • v3      |
| <b>The evaluator overstated and under-checked</b><br>Pooled metrics moved with run count, additive-only behaviour was asserted by comparing two scalars, abstention was credited on error paths, and attribution scoring was dropped.                            | Replaced with a per-scenario evaluator applying macro averages, per-run additive-only verification and an explicit replicate whitelist.                          | REPLACED        |

Disclosing defects found in our own pipeline is treated as part of the experimental result. Two of the four had fabricated data; both were corrected and re-executed, and the withdrawn figures are not reported anywhere in this deck.

# Scenario-Level Classification Results

168 live runs, 14 scenarios, 12 replicates each | Wilson score 95% confidence intervals | Expected verdicts pre-registered before execution | Source: EVALUATION\_V4.json

| SCENARIO                    | EXPECTED | PRE-FROZEN BASE RULE | REVISED RULE v3 | 95% CI (v3) |              |
|-----------------------------|----------|----------------------|-----------------|-------------|--------------|
| A1 rogue grandmaster        | ATTACK   | 12/12                | 12/12           | 0.757-1.000 |              |
| A2 Sync spoofing            | ATTACK   | 12/12                | 12/12           | 0.757-1.000 |              |
| A3 replay                   | ATTACK   | 12/12                | 12/12           | 0.757-1.000 |              |
| A5 DoS flooding             | ATTACK   | 12/12                | 12/12           | 0.757-1.000 |              |
| A8 rogue boundary clock     | ATTACK   | 12/12                | 12/12           | 0.757-1.000 |              |
| C1 selective interception   | ATTACK   | 0/12                 | 11/12           | 0.646-0.985 | gap closed   |
| C2 malformed frames         | ATTACK   | 0/12                 | 12/12           | 0.757-1.000 | gap closed   |
| C3 whole-second abuse       | ATTACK   | 0/12                 | 12/12           | 0.757-1.000 | gap closed   |
| baseline                    | BENIGN   | 12/12                | 12/12           | 0.757-1.000 |              |
| B2 planned GM changeover    | BENIGN   | 12/12                | 12/12           | 0.757-1.000 |              |
| B3 congestion               | BENIGN   | 11/12                | 11/12           | 0.646-0.985 | 1 abstention |
| B7 topology reconfiguration | BENIGN   | 12/12                | 12/12           | 0.757-1.000 |              |
| B_bc_replacement            | BENIGN   | 0/12                 | 0/12            | 0.000-0.242 | open failure |
| B_unplanned_failover        | UNKNOWN  | 12/12                | 12/12           | 0.757-1.000 |              |

Per-scenario recall is the primary metric. Three coverage gaps — interception, malformed frames and whole-second abuse — moved from 0/12 to 11/12, 12/12 and 12/12 with no change in benign performance. One benign scenario remains an open failure.

# Aggregate Classification Performance

Macro averages over per-scenario correctness rates, 168 live runs | Attack sensitivity over 8 attack scenarios, specificity over 5 benign scenarios, abstention over 1 ambiguous scenario | Source: EVALUATION\_V4.json

0.990

Macro attack sensitivity

revised rule v3 · pre-frozen base rule  
0.625 on the same captures

0.783

Macro benign specificity

identical for both rules, so the  
revision introduced no regression

1.000

Abstention correctness

ambiguous unplanned failover  
returned UNKNOWN on 12 of 12  
runs

0.875

Fault attribution accuracy

84 of 96 attack runs had the specific  
fault correctly named

WHAT IMPROVED

The revised rule closed three coverage gaps on the same captures: selective interception 0/12 to 11/12, malformed frames 0/12 to 12/12, and whole-second abuse 0/12 to 12/12. Benign performance is unchanged, and per-run verification confirms zero regressions — no run that the base rule called ATTACK was downgraded.

WHY SPECIFICITY IS 0.783 AND NOT 1.000

Two benign scenarios miss. B\_bc\_replacement produces 12 false ATTACK calls and is the dominant cause. B3 congestion contributes one UNKNOWN. Excluding the boundary-clock replacement, benign classification is 47 of 48, or 0.979 — so the loss is one identified systematic gap plus one cautious abstention, not diffuse error.

Reporting integrity

The 0.990 figure belongs to the post-observation revised rule. The pre-frozen base rule scored 0.625 on the same captures. Both are shown so the improvement is not mistaken for independent pre-registered validation.

Headline position: strong attack detection with one identified and explainable benign failure, an honest abstention on the genuinely undecidable case, and correct fault attribution on seven of eight attack scenarios.

# Rule-Based and Machine-Learning Classification Compared

Worklet objective 5 requires comparison against traditional fault management | Session-disjoint split: trained on replicates 1–8, tested on replicates 9–12 (56 held-out runs) | Hyperparameters and split fixed before scoring

ARM A • RULE-BASED

Frozen decision rule

Inputs: 56 extracted packet fields plus provisioned operator context. No training. SHA-256 frozen before the campaign; v3 is a disclosed post-campaign revision.

ARM B • MACHINE LEARNING

Random Forest + Isolation Forest

90 trees, depth 6, balanced classes, seed 1588, with an Isolation Forest novelty layer for abstention. Inputs: ptp4l servo telemetry only — offset, path delay and delay variation.

CONTROL • ALWAYS-BENIGN

Constant predictor

Returns BENIGN for every run. Included so that any apparent ML skill can be tested against the trivial baseline rather than assumed.

| METRIC ON 56 HELD-OUT RUNS | ARM A — RULE  | ARM B — ML    | ALWAYS-BENIGN |
|----------------------------|---------------|---------------|---------------|
| Exact accuracy             | 51/56 (0.911) | 24/56 (0.429) | 20/56 (0.357) |
| Attack sensitivity         | 0.969 (31/32) | 0.156 (5/32)  | 0.000 (0/32)  |
| Benign specificity         | 0.800 (16/20) | 0.950 (19/20) | 1.000 (20/20) |
| Correct abstention         | 1.000 (4/4)   | 0.000 (0/4)   | 0.000 (0/4)   |

ARM B IS NOT DISTINGUISHABLE FROM THE CONSTANT PREDICTOR

ARM B returns BENIGN on 47 of 56 runs and improves on the constant predictor by four runs. A paired exact McNemar test gives  $p = 0.219$ , so the difference is not significant at this sample size. Both score 4/4 on the boundary-clock replacement, so that result is not a unique learned capability.

NO MEASURED COMBINATION EXCEEDS THE RULE ARM

Combining the arms was evaluated rather than assumed. Logical OR reaches 0.893, rule-first with ML fallback 0.839, and consensus-else-abstain 0.429. All fall below ARM A at 0.911, so no combination is recommended on this evidence.

Scope condition

ARM B operated on 6 of 28 configured features because GNSS, SyncE, oscillator and protocol-legality telemetry do not exist in this testbed. This is a result about the available observability, not a general verdict on machine learning.

# Observability Constraints Behind the ML Result

Feature availability derived from observed linuxptp servo provenance | Schema defaults and synthetic parser fields invalidated to NaN before fitting | Impurity-based feature importances from the fitted Random Forest

The machine-learning arm underperformed because of what this testbed can observe, not because the method is unsuited to the problem. Three measured constraints account for the result.

01

## Only 6 of 28 configured features carried data

GNSS status, SyncE quality, oscillator error and every protocol-legality feature are absent from ptp4l servo logs. The model could key only on offset, path-delay and delay-variation magnitude — and one of the six, holdover rate, had zero importance, so five features carried all the signal.

02

## Two attacks starve the very telemetry the model needs

Servo samples average 88.0 per baseline run, falling to 44.5 for selective interception and 28.0 for whole-second abuse. An attack that suppresses timing messages also suppresses the evidence a telemetry-based detector depends on. The rule arm is unaffected because it reads packet legality and rate deficit directly.

03

## Importances indicate testbed-servo artefact risk

Ranked importances are offset\_abs\_max 0.268, path\_delay\_mean 0.262, offset\_std 0.208, pdv\_std 0.139 and offset\_mean 0.124 — all timing magnitude and dispersion, none protocol semantics. With a single testbed, domain shift cannot be separated from intrinsic class overlap.

**Evidence-based conclusion: on this telemetry the rule arm is the correct operational choice. Starvation alone does not explain the result — ARM B also fails on attacks where servo telemetry is intact — so feature poverty and class overlap are the primary causes. A fair re-test requires richer management-plane telemetry, not a different classifier.**

**This is an argument for retaining a standards-aware rule layer and for improving observability before granting a learned model authority over recovery. It is not a claim that machine learning is inherently unsuited to S-plane timing security.**

# Measured Detection Boundary of the Interception Detector

Rate-starvation clause fires when the delivered message rate falls below half the rate the source itself declares | Blackhole fraction drawn per replicate from randparams.py | Ratios recomputed from the per-run deep extracts

Selective interception is the one attack scenario that is not detected on every replicate. The miss is precisely located rather than described as residual error.

| CONFIGURED BLACKHOLE FRACTION OF THE POST-WARM-UP INJECTION WINDOW | REPLICATES | DELIVERED / DECLARED RATE | OUTCOME |
|--------------------------------------------------------------------|------------|---------------------------|---------|
| 75 %                                                               | n = 6      | 0.3665                    | CAUGHT  |
| 70 %                                                               | n = 3      | 0.4137                    | CAUGHT  |
| 65 %                                                               | n = 2      | 0.4581 / 0.4619           | CAUGHT  |
| 60 %                                                               | n = 1      | 0.5049                    | MISSED  |

Detection threshold · delivered / declared < 0.500      At 60 % the ratio is 0.5049 — 0.0049 absolute, or 0.98 % relative, above the threshold. The attack evades by a margin of five parts in a thousand.

WHAT WAS MEASURED AND WHAT WAS INTERPOLATED

Measured: 65 % is caught, 60 % is missed. The tested bracket is therefore 60–65 %. Linear interpolation of the delivered-to-declared ratio across that bracket places the crossing at approximately 60.5 %. That crossing is interpolated, not a directly tested point.

WHY THE OBSERVED RATE DEFICIT IS SMALLER THAN THE BLACKHOLE FRACTION

The blackhole covers the stated fraction of the window after an eight-second warm-up, while the measured window also contains that warm-up and the restore tail. A 75 % blackhole therefore yields a ratio of 0.37 rather than 0.25.

Sampling limit      Levels were drawn at random rather than swept, so replicate counts are uneven and the decisive missed level rests on a single run. The 55 % level offered by the design was never drawn; window arithmetic predicts a ratio near 0.57, which would also have evaded.

# Experimentally Verified Limitations

Each limitation is stated with the evidence that established it, the technical reason, and its implication | All figures recomputed from the 168-run corrected campaign

| OBSERVED LIMITATION                                           | EVIDENCE        | TECHNICAL REASON                                                                                                                                                                       | IMPLICATION                                                                                                                |
|---------------------------------------------------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Planned boundary-clock replacement is classified as an attack | 0/12 runs       | The provisioned context holds a single expected boundary-clock identity, so a legitimate replacement relay is indistinguishable from an unauthorised one.                              | Produces 12 false ATTACK calls and dominates the 0.783 specificity. Excluding it, benign classification is 47/48 = 0.979.  |
| Sustained interception below roughly 60 % evades detection    | 1 of 12 C1 runs | The rate-starvation clause fires below a 0.500 delivered-to-declared ratio; a 60 % blackhole yields 0.5049 and stays above it.                                                         | An attacker who paces interception under the threshold is not detected by a rate-based clause alone.                       |
| Coherent GNSS spoofing is not establishable here              | testbed limit   | Downstream PTP telemetry cannot separate a perfectly-mimicking spoof from a healthy clock; receiver RF observables, multi-antenna geometry or an independent time anchor are required. | No claim is made about GNSS spoof detection. The limit is on PTP-layer features, not on software in general.               |
| Held-out evidence is small and partly incomplete              | 4 per scenario  | Only four held-out replicates exist per scenario, and 48 of 168 runs lack pmc management telemetry.                                                                                    | Confidence intervals are wide by construction, and the ML arm sees 6 of 28 configured features.                            |
| Replicate metadata is internally inconsistent                 | 36 runs         | Thirty-six C-series context files record replicate 901 rather than their true replicate number.                                                                                        | The evaluation uses the replicate encoded in the directory name. This must be corrected before any fresh validation claim. |

Each limitation is an experimental finding with a located cause, not a residual uncertainty. Four of the five have a defined software or hardware route to resolution, set out on the following slides.

# Delivery Status and Objectives Not Yet Addressed

Status assessed against the original worklet milestone plan | Objective numbering follows the worklet definition reproduced on the opening slide

DONE PARTIAL NOT STARTED

|     |                                                                                                                      |             |
|-----|----------------------------------------------------------------------------------------------------------------------|-------------|
| M1  | <b>Problem study and literature survey</b><br>Standards baseline, reviewed source set, documented gap                | DONE        |
| M1  | <b>System architecture design</b><br>Data flow defined: capture, extract, decide                                     | DONE        |
| M2  | <b>KPI and fault taxonomy</b><br>144-parameter × 16-fault matrix; profile configuration traced to standards          | DONE        |
| M3  | <b>Digital twin and fault injection</b><br>Six-node testbed and live injection validated; digital twin not exercised | PARTIAL     |
| M4  | <b>AI model development</b><br>Random Forest and Isolation Forest evaluated; not usable on available telemetry       | PARTIAL     |
| M5  | <b>Healing logic integration</b><br>Decision engine evaluated; closed-loop healing not validated                     | PARTIAL     |
| M6a | <b>Testing and evaluation</b><br>168-run campaign plus rule-versus-ML comparison                                     | DONE        |
| M6b | <b>Paper and IP draft</b><br>Publication and patent disclosure                                                       | NOT STARTED |

WORKLET OBJECTIVES NOT YET ADDRESSED

1

**Fault prediction**  
Prediction before service disruption was not attempted in this campaign.

2

**Automated corrective action**  
No corrective action was executed or measured. ISOLATE, HOLDOVER and ESCALATE are verdict outputs, not actions taken.

3

**Digital-twin validation**  
The digital twin was not exercised; fault injection ran directly on the testbed.

4

**Outcome metrics**  
No MTTR, recovery-time, availability or downtime figure was measured on this testbed.

**Next gate**      The missing objectives must be executed and their recovery outcomes measured before any self-healing claim is made. The work reported here establishes detection and classification, which is the prerequisite for them.

# Remaining Work: Software and Hardware

Software items are actionable on the existing testbed | Hardware items require equipment not present in a software-only environment | Each item maps to a limitation stated on the preceding slides

## SOFTWARE — ACTIONABLE IMMEDIATELY

- Provisioned replacement context**  
Extend the operator context to carry approved replacement boundary-clock identities, then re-freeze and re-run. Closes the 0/12 benign failure and is expected to raise specificity toward 0.979.
- Finer interception sweep**  
Test levels between 55 % and 65 % as a sweep rather than a random draw, to replace the interpolated 60.5 % crossing with a measured one.
- Continuity-based detection**  
Add a clause that tracks message-arrival continuity alongside average rate, so that paced or bursty interception below the rate threshold is still detected.
- Richer telemetry for the ML arm**  
Supply management-plane telemetry in addition to servo logs and repeat the comparison, so the ML arm is evaluated on adequate observability.
- Metadata and re-freeze**  
Correct the 36 inconsistent replicate identifiers, re-hash the revised rule, and re-run the campaign for an independently pre-registered result.

## HARDWARE — REQUIRES EQUIPMENT

- GNSS receiver**  
Unlocks GNSS spoofing (A6), GNSS jamming (A7) and genuine holdover (B1) — three of the five currently unreachable classes.
- Second physical machine**  
Provides a second independent oscillator, making oscillator drift (B6) reproducible. Clocks in one host cannot genuinely drift apart.
- Hardware-timestamping NIC**  
Replaces microsecond-scale software timestamping with nanosecond-scale hardware timestamping, enabling credible physical-impact measurement for the delay attack (A4) and static asymmetry (B5).
- SyncE-capable PHY**  
Unlocks SyncE and EEC degradation (B4). SyncE is a physical-layer function and veth interfaces have no PHY, so ESMC quality signalling cannot exist in this testbed.

Five of sixteen catalogued classes require hardware; three more have software detection logic but need hardware for credible physical-impact measurement. The software items above are unblocked and can begin on the existing testbed.

# Open Fronthaul S-Plane Timing Security

## 1 Scope investigated

Discrimination of malicious timing behaviour from benign timing faults on the Open Fronthaul S-plane, between O-DU and O-RU.

## 2 Implementation

Six linuxptp 4.0 daemons under the ITU-T G.8275.1 telecom profile across two bridged segments, with faults produced on the live packet path.

## 3 Dataset produced

168 live runs across 14 scenarios at 12 randomised replicates each; 1,451,909 packet records extracted at 56 fields per packet, with unobservable fields recorded EMPTY.

## 4 Taxonomy evaluated

Eight attack scenarios, five benign scenarios and one deliberately ambiguous case, drawn from a sixteen-class catalogue of which eight were executable in software.

## 5 Classification performance

Macro attack sensitivity 0.990 for the revised rule against 0.625 pre-frozen; specificity 0.783, or 0.979 excluding the identified benign failure; abstention 12/12; attribution 0.875.

## 6 Comparison outcome

On 56 held-out runs the rule arm reached 0.911 against 0.429 for the ML arm and 0.357 for a constant predictor; the ML arm is not statistically distinguishable from that control on the available telemetry.

## 7 Limits identified

One systematic benign failure with a defined fix, an interception evasion band near 60 %, and five fault classes requiring hardware.

## 8 Immediate next steps

Extend the provisioned-replacement context and re-freeze; sweep interception levels; add continuity-based detection; re-run the ML arm with management-plane telemetry.

# Thank you

Tanmaya Kumar · Raghu Ram K · Munipalle Jaswanth Kumar

Status: S-plane detection and classification evaluated. Prediction, digital twin and automated healing not yet executed.